Day 31 HW

Thursday, May 11, 2023 6:01 PM

HOMEWORK Day 31 (5 Problems)

Polynomials	5 Problems

8-87.
For each of the following sets of numbers, find the equation of a function that has these numbers as roots. Show your work. Homework Help

a. $-3+i \; and \; -3-i$

b. $5+\sqrt{3} \; and \; 5-\sqrt{3}$

c. $-2, \sqrt{7}, and \; -\sqrt{7}$

d. $4, -3+i, \; and \; -3-i$

8-122.
Without actually multiplying, decide which of the following polynomials could be the product of $(x-2)(x+3)(x-5)$. Justify your choice. Homework Help

a. $x^3-4x^2-11x-5$

b. $2x^3-4x^2-11x+30$

c. $x^3-4x^2-11x+30$

d. $2x^3-4x^2-11x-5$

8-139.
Spud has a problem. He knows that the solutions for a quadratic equation are $x = 3 + 4i$ *and* $x = 3 - 4i$, but in order to get credit for the problem he was supposed to have written down the original equation. Unfortunately, he lost the paper with the original equation on it. Luckily, his friends are full of advice. Homework Help

Just remember when we made polynomials. If you wanted 7 *and 4 to be the answers, you just used* (x−7)(x−4). *So you just do* x *minus the first one times* x *minus the other*." Use $\left(x - (3 + 4i)\right)\left(x - (3 - 4i)\right)$to find the quadratic equation.

Hugo says, "*No, no, no. You can do it that way, but that's too complicated. I think you just start with* $x = 3 + 4i$ *and work backward. So* $x - 3 = 4i$*, then, hmmm... Yeah, that'll work*." Try Hugo's method.

Whose method do you think Spud should use? Explain your choice.

8-140.
So far you have been able to extend the rules for real numbers to add, subtract, and multiply complex numbers, but what about dividing? Can you use what you know about real numbers to divide one complex number by another? In other words, if a problem looks like this:

$$\frac{3 + 2i}{-4 + 7i}$$

What needs to be done to get an answer in the form of a single complex number, $a + bi$?

Natalio had an idea. He said, "*I'll bet we can use the conjugate!*"

"*How*?" asked Ricki.

"*Well, it's a fraction. Can't we multiply the numerator and denominator by the same number*?" Natalio replied. Homework Help

a. Natalio was not very clear in his explanation. Show Ricki what he meant they should do.

b. Using Natalio's ideas you probably still came up with a fraction in part (a), but the denominator should be a whole number. T o write a complex number such as$\frac{c+di}{m}$ in the form $a + bi$, just use the distributive property to rewrite the result as $\frac{c}{m} + \frac{d}{m}i$. Rewrite your result for part (a) in this form.

8-141.

Use the method developed in problem 8-140 to do the following division problems. Homework Help

a. $\dfrac{2-5i}{1-2i}$

b. $(-3+i) \div (2+3i)$